

OBSERVER

● DEPLOYED & ACTIVE

Autonomous Host & Container SRE Monitoring Operator

DATE: 18 SEP 2026

Observer is Volcano's centralized monitoring and incident response engine. Deployed to monitor 29+ production containers, ingest Loki log streams in real-time, and route human-readable digests to Telegram.

< 60s

Incident Alert Latency

0 False Alarms

Grafana Alert Noise Elimination

24/7

Autonomous Health Polling

TECHNICAL CHANGES & HIGHLIGHTS

■ Real-Time Loki Error Stream Poller

Implemented background async monitor scanning container logs for exceptions and panics with a 5-minute per-service debounce.

■ Automated Daily Performance Reports

Automated 09:00 AM server reports aggregating NVMe disk capacity, RAM consumption, CPU load, and 24h error frequency.

■ Telegram Channel Routing & Sanitized HTML

Replaced raw markdown with sanitized Telegram HTML and category routing (#alerts, #deploys, #reports).

■ Live System Status API (GET /status)

Added endpoint exposing real-time container health and deploy history for portfolio integration.

BUSINESS & OPERATIONAL IMPACT

Zero-Downtime Awareness: Immediate push visibility into PostgreSQL connection exhaustion, container crashes, and API exceptions across all apps before users report them.

High Signal-to-Noise Ratio: Eliminated alert fatigue by replacing raw label dumps with structured diagnostics and debounced summaries.

Automated DevOps Auditing: Every code deployment to Volcano now automatically posts confirmation and audit trails directly to mobile.

PORTFOLIO & CONTROL CENTER

● PRODUCTION LIVE

Unified Multi-Dashboard Portal at mileshillary.com/admin

DATE: 18 SEP 2026

mileshillary.com was transformed into both a dynamic showcase of active projects and a unified single-pane-of-glass administrative control center hosting all backend engineering dashboards.

1 URL

Unified Admin Access Point

5 Dashboards

Consolidated in Multi-Tab UI

100% Real-Time

Dynamic Project Health Sync

TECHNICAL CHANGES & HIGHLIGHTS

■ Unified Admin Portal (/admin)

Built a tabbed command center consolidating Grafana Metrics, Clearbox Supabase Studio, Horizon Studio, Crucible QA, and Observer Ops into a single URL.

■ Dynamic Status Engine

Connected homepage status indicator dots to Observer's live status API, dynamically reflecting live, updating, and archived states.

■ Nginx Internal Reverse Proxy

Configured Nginx dynamic DNS resolution to securely proxy API status and report triggers to internal Docker containers on the observability network.

■ Deploy Notification Hook

Added automated CI/CD webhooks into the deployment workflow to notify Observer upon every successful site build.

BUSINESS & OPERATIONAL IMPACT

Operational Consolidation: Eliminated fragmented port-hopping across local IPs and remote endpoints; manage databases, metrics, and QA in one browser tab.

Accurate Public Representation: Visitors to mileshillary.com see verified, real-time live statuses for active production products (Volc, Clearbox, Horizon).

Instant Actionability: One-click 'Telegram Report' trigger right from the web UI to inspect server diagnostics on demand.

VOLC AI GYM COACH

● APP STORE LIVE

iOS Computer Vision Coaching & Backend Analytics Engine

DATE: 18 SEP 2026

Volc is an AI-powered workout tracking and computer vision coaching platform consisting of a live iOS application, high-throughput backend API, and landing page with integrated telemetry analytics.

Live iOS App

App Store ID: 6751469055

v2.0 Backend

FastAPI + Dockerized Stack

1-Click CTA

Platform-Aware App Store Routing

TECHNICAL CHANGES & HIGHLIGHTS

■ Unified Multi-Container Stack

Integrated landing page website (volc-website:3004) and backend API (supreme-octo-doodle-api:8002) under unified Docker Compose with healthchecks.

■ Admin Analytics Dashboard

Engineered Highcharts visual analytics dashboard with hourly backend aggregation for workout sessions and active users.

■ Platform-Aware Landing Page

Updated web client with native App Store badges for iOS users and inline email capture for Android/web visitors.

■ Observer Health & Error Monitoring

Integrated Volc backend and frontend into Observer's automated 24h error log parser and live status engine.

BUSINESS & OPERATIONAL IMPACT

Conversion Rate Optimization: Dynamic platform detection routes iOS visitors directly to the App Store while capturing Android/desktop leads for future rollouts.

Session Reliability: Continuous monitoring of workout telemetry endpoints ensures uninterrupted video analysis and exercise logging for gym users.

User Retention Visibility: Real-time Highcharts analytics provide daily and hourly insights into workout completion and repeat usage patterns.

CLEARBOX MEDICINE API

● PRODUCTION LIVE

Self-Hosted Supabase, Pgvector & Biomedical Telemetry

DATE: 18 SEP 2026

Clearbox is a biomedical knowledge retrieval and clinical adherence platform running a complete self-hosted Supabase infrastructure (PostgreSQL 16, pgvector, PostgREST, Studio, and Kong Gateway).

Pgvector PG16

Biomedical Embeddings DB

Port 8009

Supabase Studio Console

4 Services

Complete Self-Hosted DB Stack

TECHNICAL CHANGES & HIGHLIGHTS

■ Self-Hosted Supabase Stack

Deployed PostgreSQL with pgvector, PostgREST (:3006), GoTrue auth, and Supabase Studio (:8009) on the internal Volcano network.

■ Embedded Studio in Unified Admin

Integrated Clearbox Supabase Studio directly into mileshillary.com/admin for instant database table and embedding management.

■ Clinical Telemetry & Analytics Aggregator

Wired analytical join aggregators (dmd_ampp) for tracking patient adherence, retention metrics, and dosage schedules.

■ LogQL Diagnostic Filtering

Hardened Observer error queries to distinguish actual database exceptions from telemetry log payloads containing error substrings.

BUSINESS & OPERATIONAL IMPACT

Zero Third-Party Cloud DB Cost: Running a complete enterprise Supabase + pgvector stack locally on Volcano eliminates recurring cloud database expenses.

Direct Data Governance: Full control over biomedical embedding tables and patient telemetry data with zero data leaving the private server infrastructure.

Rapid Incident Resolution: Real-time Loki error alerts instantly catch schema mismatches and PostgREST JWT authentication failures before users notice.

HORIZON & VOLCANO CI/CD

Algorithmic Token Fund & Cloudflare Zero Trust CI/CD

● ACTIVE DEPLOYMENT

DATE: 18 SEP 2026

Horizon runs simulation token trading on Solana paired with automated server deployment pipelines across all repositories via Cloudflare Zero Trust Tunnels.

29 Containers Active Monitored Fleet	< 45s Git Push to Live Deployment	Zero Trust Cloudflare HTTP/2 Tunnels
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TECHNICAL CHANGES & HIGHLIGHTS

■ Capital Rotation Engine Refactored Horizon's wallet manager to automatically deduct reinvested capital from the cash pool upon strategy rotations.
■ Portfolio Net % Valuation Corrected portfolio ROI calculations to measure returns against actual invested capital rather than theoretical limits.
■ Standardized GitHub Actions CI/CD Equipped all project workflows with Cloudflare SSH proxy commands and automated Telegram deploy webhooks.
■ Zero Trust Tunnel Optimization Updated cloudflared tunnel sidecar parameters to resolve websocket/HTTP2 handshake drops for remote tunnels.

BUSINESS & OPERATIONAL IMPACT

Algorithmic Capital Accuracy: Precise cash pool and ROI accounting prevents simulation capital drift during high-frequency trading.
Sub-45s Developer Velocity: Any commit pushed to main across any project tests, builds, and deploys to production with zero manual SSH steps.
Foundation for Autonomous AI SRE: Established the network topology and API layer required for Antigravity AI agents to diagnose and repair codebases.